**10 (a)** State Wien's displacement law.

.....

..... [1]

**(b)** Fig. 10.1 shows the wavelength distributions of electromagnetic radiation emitted by two stars A and B.

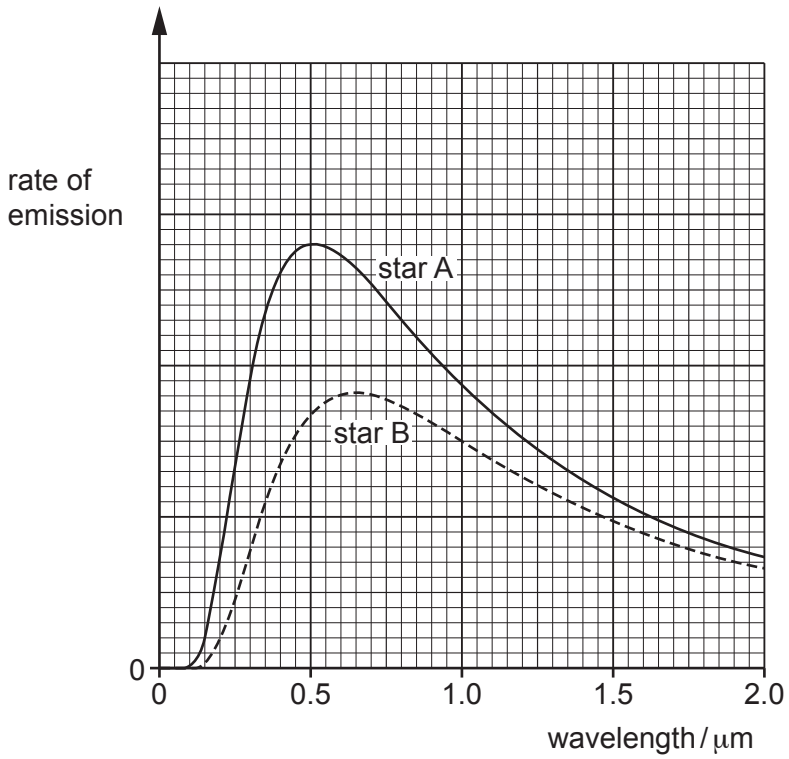



**Fig. 10.1**

The surface temperature of star A is known to be 5800 K.

**(i)** Determine the surface temperature of star B.

surface temperature = ..... K [2]

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**(ii)** Star B appears less bright than star A when viewed from the Earth.

Use Fig. 10.1 to suggest, with a reason, how else the physical appearance of star B compares with that of star A.

.....

.....

..... [2]

**(c)** The lines in Fig. 10.1 have been corrected for redshift.

**(i)** State what is meant by redshift.

.....

.....

..... [2]

**(ii)** Explain how cosmologists are able to determine that light from a distant star has undergone redshift.

.....

.....

..... [2]

[Total: 9]

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